

Shashank Yadav Kalki

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PROFESSIONAL SUMMARY

Detail-oriented Data Analyst with an M.Sc. in Mathematical Modelling & Machine Learning from University College Cork. Experienced in SQL, Python, Excel, Tableau, and statistical analysis for data wrangling, dashboard development, business intelligence, and reporting. Skilled in transforming structured and unstructured data into actionable business insights through data cleaning, validation, visualization, and analytical storytelling. Strong foundation in machine learning, statistical modelling, and stakeholder-focused data-driven decision making.

TECHNICAL SKILLS

Programming: Python (NumPy, pandas, Scikit-learn), SQL, R
Business Intelligence: Tableau, Excel, Pivot Tables, Power Query, Data Visualization, Dashboard Development
Data Analytics: Data Wrangling, Data Cleaning, Data Validation, Exploratory Data Analysis, Statistical Analysis, Business Intelligence, Business Analytics, KPI Reporting, Reporting
Database: SQL Queries, Joins, Aggregations, Common Table Expressions (CTEs), Window Functions, Data Warehousing Concepts
Machine Learning: Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Time Series Analysis
Mathematics: Statistical Modelling, Statistical Inference, Optimization, Numerical Simulation, Probability
Tools: FastAPI, Git/GitHub, Excel, Google Sheets, Matplotlib, Seaborn

RESEARCH EXPERIENCE

Machine Learning Research Assistant (MSc Dissertation) May 2025 – Sept 2025
University College Cork, Ireland | Supervisor: Dr. Andrew Flynn

- Designed and executed computational experiments involving large experimental datasets to investigate reliability and emergent behaviour in reservoir computing systems.
- Developed reproducible Python workflows for data preprocessing, statistical analysis, visualization, validation, and model evaluation.
- Performed exploratory data analysis, quantitative analysis, and statistical modelling to identify patterns influencing prediction reliability.
- Produced analytical reports and visualizations to communicate findings to academic stakeholders.
- Authored a 50-page MSc dissertation titled *Confabulation Dynamics in a Reservoir Computer*.

PROJECTS

HR Workforce Analytics Dashboard June 2026
Power BI | SQL | Excel | Power Query

- Designed an interactive Power BI dashboard to analyse workforce metrics including employee headcount, attrition, department-wise performance, hiring trends, and key HR KPIs.
- Performed SQL-based data wrangling using joins, aggregations, and Common Table Expressions (CTEs) to prepare reporting datasets from multiple data sources.
- Cleaned, transformed, and validated HR datasets using Excel Power Query to improve data quality and reporting accuracy.
- Built KPI scorecards and drill-through dashboards enabling business stakeholders to monitor workforce performance and recruitment trends.
- Automated reporting workflows, reducing manual effort and improving decision-making through interactive business intelligence visualizations.

Sales & Marketing Performance Dashboard May 2026
Tableau | SQL | Excel

- Developed an interactive Tableau dashboard analysing sales performance, customer acquisition, regional trends, and marketing campaign effectiveness.
- Wrote SQL queries to clean, transform, and aggregate structured business data for reporting and analytical insights.
- Created executive KPI dashboards with drill-down visualizations to identify high-performing products, customers, and sales regions.
- Performed data validation, quality assurance, and testing to ensure dashboard accuracy before dashboard deployment.
- Generated actionable business insights supporting sales forecasting, marketing strategy, and stakeholder decision-making.

Heart Disease Prediction – End-to-End ML Pipeline Jan 2025 – Mar 2025
Python | Scikit-learn | SQL | FastAPI

- Built an end-to-end machine learning pipeline including data preprocessing, data cleaning, feature engineering, model training, hyperparameter tuning, and evaluation.
- Performed exploratory data analysis and statistical analysis to identify trends, missing values, and feature importance within structured healthcare datasets.
- Compared supervised learning algorithms using Accuracy, Precision, Recall, F1-score, and ROC-AUC to evaluate predictive performance.
- Performed feature selection, data validation, and model evaluation to improve robustness, reliability, and prediction accuracy.
- Developed FastAPI endpoints for model inference and integrated analytical outputs into a reusable prediction service.

Human Activity Recognition using Wearable Sensor Data Oct 2024 – Jan 2025
Python | PCA | Time Series | Scikit-learn

- Processed wearable sensor datasets through data cleaning, preprocessing, and feature engineering for human activity recognition.
- Applied Principal Component Analysis (PCA) to reduce dimensionality while preserving predictive information and improving computational efficiency.
- Performed exploratory data analysis, statistical validation, and classification model evaluation across multiple machine learning algorithms.
- Analysed prediction errors and model performance to improve model generalization and classification accuracy.
- Produced analytical reports and data visualizations to communicate findings and support data-driven decision-making.

EDUCATION

University College Cork, Ireland Cork, Ireland
M.Sc. Mathematical Modelling & Machine Learning (Second Class Honours Grade I) Sept 2024 – Mar 2026
Mahatma Gandhi Institute of Technology Hyderabad, India
B.Tech Computer Science (Data Science) 2020 – 2024

CERTIFICATIONS

- Data Science for Engineers – IIT Madras
- Natural Language Processing – IIT Kharagpur
- Data Analysis with Python – FreeCodeCamp
- Social Network Analysis – IIT Ropar
- Prompt Engineering with OpenAI – Columbia University
- Generative AI Masterclass – Udemy